

MODELLING AND PERFORMANCE ANALYSIS OF TRANSMISSION OF A GO-KART

*A Mini-Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of*

**BACHELOR OF TECHNOLOGY
IN
MECHANICAL ENGINEERING**

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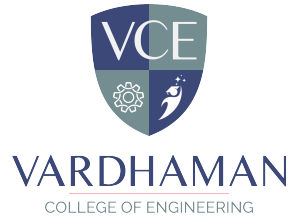
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CERTIFICATE

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Abstract

A go-kart is a small and low-height vehicle designed for recreational activities and entry-level motorsport racing. A go-kart is typically a light-weight chassis with a compact-sized engine/motor and a simple steering and brake system along with four wheels for optimal stability on the ground. A go-kart is recognized for its high maneuverability and quick acceleration over short distances, making it an ideal platform for learning the basics of driving and racing.

The design of a go-kart is aimed at creating a compact-sized, stable, and high-performance vehicle that ensures safety and efficiency in driving and racing. The drivetrain system is designed using a chain and sprocket system for optimal torque and acceleration performance. Perform calculations to select the optimal gear ratio to ensure optimal power delivery from the motor to the wheels. The objective of the work is to design and model and analyze the powertrain of a Go-Kart.

Keywords: Go-kart; transmission; gear ratio; performance parameters; steering system; braking system

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Abbreviations

Abbreviation	Description
CAD	Computer Aided Drawing
kW	KiloWatt
RPM	Rotations per minute
v	Volts
P	Power
T	Torque
N	Speed
N	Newton
m	Meter
N-M	Newton meter
km/h	Kilometer per hour
m/s ²	Meter per second square

CHAPTER 1

Introduction

1.1 Introduction

Go-kart is a light-weight racing vehicle used for driver training, racing, and engineering projects. The transmission plays a vital role in electric go-kart racing. The transmission is used to transmit power to the rear wheels of the electric motor.

This work focuses on the modelling and performance analysis of the transmission system of an electric go-kart. The aim of the project is to understand the mechanism of power transmission from the motor to the kart's drivetrain, and to understand the effect of the transmission system on the performance characteristics of the kart. The major systems of the kart, such as the steering system, braking system, and the transmission system, were modeled using CAD to understand the mechanical arrangement of the kart.

Apart from the modeling, basic analytical calculations were performed to compute important parameters such as motor torque, tractive force, acceleration, and speed. These calculations are important in evaluating the performance of the drivetrain design. The research provides valuable knowledge on the design of the electric go-kart drivetrain.

1.2 Background and Motivation

Electric go-karts are increasingly being used in Indian karting races and student motorsport events because of their high efficiency, instant torque, and eco-friendliness. The performance of a racing kart is significantly dependent on its transmission system, which transmits power from the electric motor to the wheels. Indian karting circuits have short straights and a number of corners, where acceleration is a very crucial aspect.

Hence, it is important to understand and analyze the transmission system to enhance the performance of the drivetrain and obtain better racing performance. This work was initiated to model and analyze the transmission system of an electric go-kart and to know the impact of drivetrain design on the performance of the vehicle in Indian kart racing conditions.

1.3 Objectives of the Project Work

- To model a go-kart compliant with Indian karting race regulations.
- To select an appropriate drivetrain.
- To model key subsystems such as the steering system, braking system, and drivetrain using CAD software.
- To model powertrain parameters for optimum acceleration and speed.
- To perform analytical calculations to estimate motor torque, tractive force, vehicle acceleration, and top speed.

1.4 Organization of the Report

This report is organized into several chapters to systematically present the modelling and performance analysis of the transmission system of an electric go-kart. Chapter 1 introduces the project by presenting the background, motivation, objectives, and overall scope of the study.

- **Chapter 2: Literature Review**
 - This chapter reviews previous research related to electric go-kart design and transmission systems. It discusses existing studies on powertrain configuration, motor selection, drivetrain efficiency, and performance analysis of electric karts. The chapter helps identify the research gap and the need for detailed modelling and analysis of the transmission system for improved kart performance.

- **Chapter 3:** Design and Modelling of Electric Go-Kart Transmission System

- This chapter explains the design and modelling of the go-kart components used in the project. It includes the layout of the powertrain system and CAD modelling of important subsystems such as the steering system, braking system, and transmission assembly.

- **Chapter 4:** Performance Analysis of the Electric Go-Kart Transmission System

- This chapter presents the analytical calculations and performance evaluation of the transmission system. It includes calculations of torque, gear ratio, tractive force, acceleration, and vehicle speed based on the selected drivetrain configuration.

- **Chapter 5:** Results and Discussion

- This chapter presents the results and discussion, where the calculated results are analyzed to evaluate the effectiveness of the transmission design and its influence on the performance of the electric go-kart.

- **Chapter 6:** Conclusions and Recommendations

- This chapter provides the conclusions and recommendations, summarizing the key findings of the study and suggesting possible improvements and future work for enhancing the performance of electric go-karts.

CHAPTER 2

Literature Survey

2.1 Introduction

This chapter discusses a review of previous literature concerning the design and performance analysis of electric go-kart and transmission systems. The purpose of this literature survey is to understand the existing work done in electric vehicle powertrains and drivetrain systems. By studying previous research papers and technical studies, important concepts such as motor selection, transmission design, and vehicle performance evaluation can be understood. This chapter also discusses the importance of previous literature in identifying the gap and the foundation of the current work.

2.2 Review of Prior Research

Several researchers have studied the design and development of electric go-karts and their drivetrain systems. Many studies focus on the overall design of the kart including the chassis, steering system, braking system, and electric powertrain. Some researchers have analyzed the selection of electric motors and battery systems to achieve efficient power delivery and improved vehicle performance.

Previous research also discusses the use of chain-sprocket transmission systems for transferring power from the motor to the rear axle, as it provides high efficiency and simple mechanical design. In addition, several studies have carried out performance calculations to estimate parameters such as vehicle speed, acceleration, and tractive force based on motor specifications and transmission ratios. These studies provide useful insights into drivetrain design and highlight the importance of proper transmission configuration in achieving better vehicle performance.

2.3 Identified Research Gaps

Although several studies have been conducted on the design and fabrication of electric go-karts, many of them mainly focus on structural design and overall vehicle assembly. Limited research has been carried out on the detailed modelling and performance analysis of the transmission system. In particular, the influence of transmission parameters such as sprocket sizes and gear ratio on vehicle acceleration and speed has not been extensively analyzed. Furthermore, there is limited work addressing the optimization of drivetrain design for specific racing conditions such as those found in Indian karting tracks. These gaps highlight the need for further analysis and modelling of the transmission system to better understand its effect on electric go-kart performance.

2.4 Summary

Previous studies have provided useful information regarding motor selection, drivetrain configuration, and vehicle performance analysis. However, the literature review also revealed that detailed analysis of the transmission system and its effect on vehicle performance is relatively limited. Therefore, this project focuses on modelling and analyzing the transmission system of an electric go-kart to better understand its role in improving drivetrain efficiency and overall vehicle performance. The findings from this study aim to contribute to the development of more efficient and competitive electric go-kart designs.

CHAPTER 3

Design and Modelling of Electric Go-Kart Transmission System

3.1 Introduction

This chapter describes the design and modelling of the electric go-kart system used in this work. The main objective of this chapter is to explain the configuration of the drivetrain system and the modelling of important components such as the steering system, braking system, and transmission system. Proper design and modelling help in understanding the arrangement of components and the power transmission from the electric motor to the rear wheels. The modelling of the components was carried out using CAD software to visualize the structure and integration of the go-kart subsystems.

3.2 Overview of the Electric Go-Kart System

The transmission system is responsible for transferring power from the electric motor to the rear wheels. In this work, a chain-sprocket transmission system was selected due to its high efficiency, simplicity, and suitability for go-kart applications. The system consists of a motor sprocket, chain drive, rear sprocket, and rear axle. The motor sprocket is connected to the motor shaft, and the chain transmits the rotational motion to the rear sprocket mounted on the rear axle. The selection of appropriate sprocket sizes helps in achieving the desired transmission ratio, which influences the vehicle's acceleration and speed.

3.3 Methodology

The design process of transmission of the electric go-kart involves the following steps :

- Literature review
- Identification of System Requirements
- CAD Modelling of Steering, Braking and Transmission Systems
- Transmission System Design
- Selection of Sprocket Sizes
- Performance parameters Calculations
- Performance Evaluation of Electric Go-Kart

3.4 Transmission System Modelling

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The motor sprocket is connected to the motor shaft, and the chain transmits the rotational motion to the rear sprocket mounted on the rear axle. The selection of appropriate sprocket sizes helps in achieving the desired transmission ratio, which influences the vehicle's acceleration and speed.

3.5 Steering System Modelling

The steering system is an important subsystem that allows the driver to control the direction of the go-kart. In this work, the steering components such

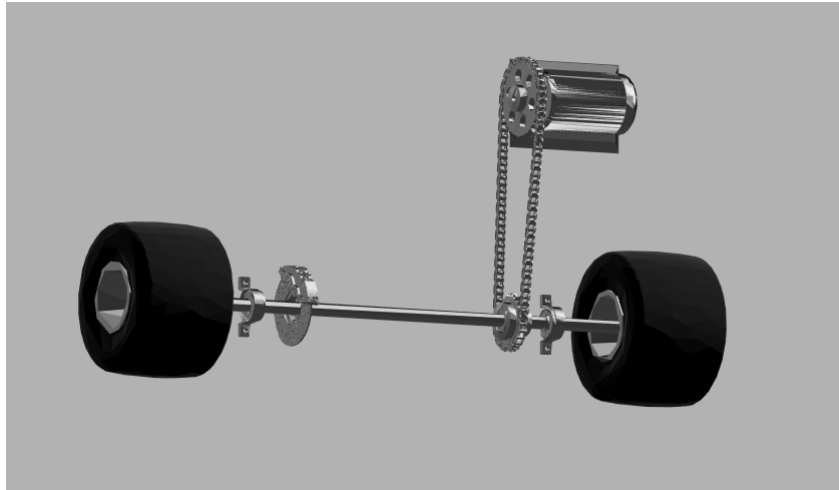


Figure 3.1: Transmimssion system

as the steering wheel, steering column, tie rods, and steering knuckles were modelled using CAD software. The steering system was designed to provide smooth directional control and maintain stability during vehicle operation.

Proper alignment and arrangement of the steering components are important to ensure accurate turning and driver control. The model ensures that the steering system can operate without interference with other subsystems such as the chassis and braking components.

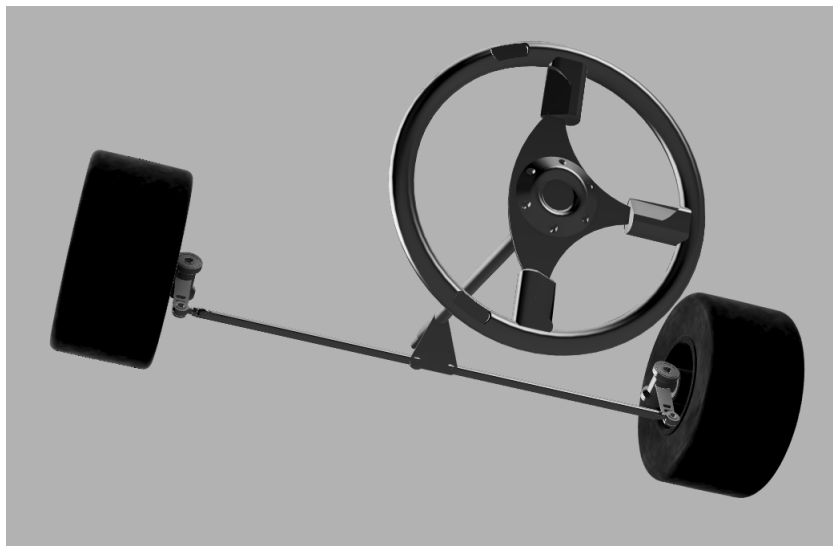


Figure 3.2: Steering system

3.6 Braking System Modelling

The braking system is essential for controlling the speed of the vehicle and ensuring safety. In this work, the braking system components including the brake disc and brake caliper mechanism were modelled using CAD software. The braking system works by applying friction between the brake pads and the brake disc, which reduces the rotational speed of the wheels and brings the vehicle to a stop.

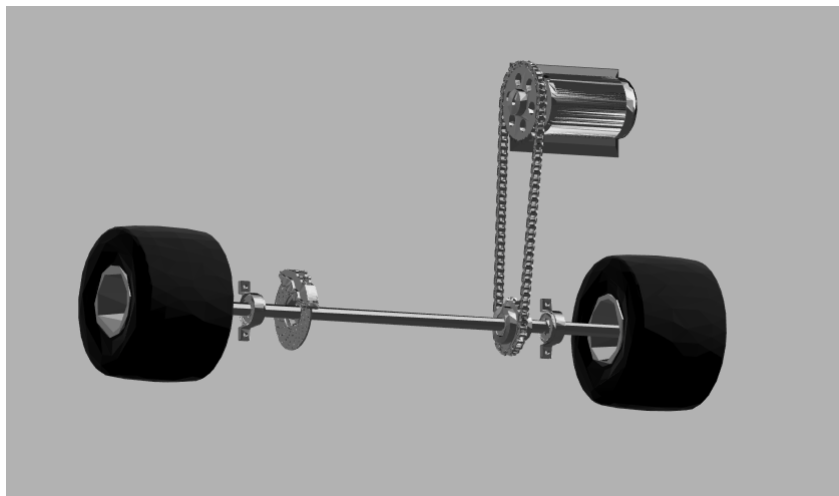


Figure 3.3: Braking system

3.7 Summary

The modelling of the key subsystems of electric go-kart system are presented. The configuration of the drivetrain system and the modelling of key subsystems such as the steering system, braking system, and transmission system were discussed. The CAD models help in understanding the arrangement and integration of the components within the go-kart. The next chapter focuses on the performance analysis of the transmission system, including the calculations used to evaluate the vehicle's performance.

CHAPTER 4

Performance Analysis of the Electric Go-Kart Transmission System

4.1 Introduction

This chapter presents the performance analysis of the transmission system of the electric go-kart. The objective of this analysis is to evaluate how effectively the transmission system transfers power from the electric motor to the rear wheels. Analytical calculations are performed to determine important parameters such as motor torque, transmission ratio, rear axle torque, tractive force, acceleration, and vehicle speed. These calculations help in understanding the influence of drivetrain design on the overall performance of the go-kart.

4.2 Motor specifications

The performance analysis is based on the specifications of the electric motor used in the go-kart.

<i>Parameter</i>	<i>Value</i>
Rated Power	6 kW
Peak Power	10 kW
Motor Speed	3000 rpm
Voltage	144 V

Table 4.1: Motor Specifications

4.3 Calculations

4.3.1 Motor Torque

Motor torque is calculated using the power equation:

$$P = \frac{2\pi NT}{60}$$

Where,

P = Power (W)

N = Motor speed (rpm)

T = Torque (N-m)

Rearranging the equation:

$$T = \frac{P \times 60}{2\pi N}$$

Substituting the values:

$$T = \frac{6000 \times 60}{2\pi \times 3000} = 19.1 \text{ N-m}$$

Motor Torque = 19.1 N-m

4.3.2 Gear Ratio

The transmission system uses a chain-sprocket mechanism.

Motor sprocket teeth = 12

Rear axle sprocket teeth = 36

$$GearRatio = \frac{\text{Rear sprocket teeth}}{\text{Motor sprocket teeth}}$$

Substituting the values:

$$GearRatio = \frac{36}{12} = 3$$

Gear Ratio = 3 : 1

4.3.3 Torque at Rear Axle

The torque at the rear axle increases due to the gear ratio.

$$T_{\text{axle}} = T_{\text{motor}} \times \text{Gear Ratio} \quad (4.1)$$

$$T_{\text{axle}} = 19.1 \times 3 = 57.3 \text{ N-m} \quad (4.2)$$

Rear Axle Torque = 57.3 Nm

4.3.4 Tractive Force

Tractive force is the force responsible for moving the vehicle forward.

Wheel radius = 0.14 m

$$F = \frac{T}{r} \quad (4.3)$$

$$F = \frac{57.3}{0.14} = 409.28 \text{ N} \quad (4.4)$$

Tractive Force = 409.28 N

4.3.5 Vehicle Speed

Wheel diameter = 0.28 m

Wheel circumference:

$$C = \pi D \quad (4.5)$$

$$C = \pi \times 0.28 = 0.879 \text{ m} \quad (4.6)$$

C = 0.879 m

Wheel RPM:

$$\text{Wheel RPM} = \frac{\text{Motor RPM}}{\text{Gear Ratio}} \quad (4.7)$$

$$\text{Wheel RPM} = \frac{3000}{3} = 1000 \text{ rpm} \quad (4.8)$$

Wheel RPM = 1000 rpm

Vehicle speed:

$$V = \frac{\text{RPM} \times C \times 60}{1000} \quad (4.9)$$

$$V = \frac{3000 \times 0.879 \times 60}{1000} = 52.74 \text{ km/h} \quad (4.10)$$

Vehicle Speed 52.74 km/h

4.3.6 Acceleration

Vehicle mass = 160 kg

Using Newton's second law:

$$a = \frac{F}{m} \quad (4.11)$$

$$a = \frac{409.28}{160} = 2.55 \text{ m/s}^2 \quad (4.12)$$

Acceleration = 2.55 m/s²

4.4 Summary

The analytical calculations show that the selected drivetrain configuration provides sufficient torque and tractive force to drive the electric go-kart effectively. The estimated vehicle speed of approximately 53 km/h and the calculated acceleration demonstrate that the chosen transmission system is suitable for electric go-kart racing applications.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the results obtained from the performance analysis of the electric go-kart transmission system. The purpose of this chapter is to analyze the calculated parameters and evaluate the effectiveness of the drivetrain design. The results are related to the project objectives, which focus on understanding how the transmission system influences the performance of the electric go-kart. This chapter first provides an overview of the obtained results and then discusses their interpretation and evaluation based on several quality factors such as environmental impact, sustainability, safety, cost, and compliance with engineering standards.

5.2 Overview of Results

The performance analysis of the electric go-kart transmission system produced several important results related to motor torque, tractive force, vehicle acceleration, and vehicle speed. The calculations were carried out considering a 6 kW electric motor operating at 3000 rpm and a chain-sprocket transmission system with 12 teeth on the motor sprocket and 36 teeth on the rear axle sprocket.

The results obtained from the analysis are summarized in the table below.

<i>Parameter</i>	<i>Value</i>
Motor Power	6 kW
Motor Speed	3000 rpm
Motor Torque	19 N-m
Gear Ratio	3 : 1
Rear Axle Torque	57 N-m
Tractive Force	407 N
Vehicle Speed	52.7 km/h
Acceleration	2.54 m/s ²

Table 5.1: Analytical results

5.3 Analysis and Interpretation

The calculated motor torque of approximately 19 N-m indicates that the electric motor provides enough torque to drive the go-kart. With the selected sprocket configuration, a 3:1 gear ratio is obtained, which increases the torque at the rear axle to about 57 N-m. This increase in torque helps generate the tractive force required to accelerate the vehicle.

The tractive force calculated at the wheels is approximately 407 N, which is sufficient to overcome the resistance forces acting on the vehicle and to move forward. The estimated vehicle speed of about 53 km/h demonstrates that the selected transmission configuration offers a balanced combination of acceleration and top speed.

In kart racing environments, especially on tracks with multiple turns, rapid acceleration is often more important than extremely high top speeds. Therefore, the selected transmission configuration is suitable for electric go-kart applications where both speed and acceleration are required for efficient racing performance.

5.4 Evaluation of Quality Factors

The results are evaluated based on several quality factors. These factors include:

- **Environment:** The use of an electric motor instead of an internal combustion engine reduces harmful emissions and noise pollution. This contributes to a cleaner and more environmentally friendly racing system.
- **Sustainability:** Electric go-karts promote sustainable mobility by reducing fuel consumption and supporting the use of renewable energy sources for charging batteries.
- **Safety:** The design includes a braking system and proper steering control to ensure safe operation. The transmission system also provides controlled power delivery with the aid of a controller, which improves vehicle stability.
- **Ethics:** The project follows standard engineering practices and ensures that the design and analysis are performed with academic integrity and responsible research methods.
- **Cost:** The chain-sprocket transmission system used in this project is simple and cost-effective compared to gearbox systems. This makes it suitable for student to participate in competitions.
- **Type:** The results obtained are analytical and design-oriented, which are appropriate for evaluating drivetrain performance in mechanical engineering projects related to electric vehicles.
- **Standards:** The design procedure follows general mechanical engineering design principles used in electric vehicle drivetrain systems.

5.5 Summary

The results obtained from the performance analysis of the electric go-kart transmission system are displayed above. The results show that the selected drivetrain configuration provides good amount of torque, tractive force, and vehicle speed for electric go-kart operation. The analysis also shows that the transmission system plays a critical role in determining the overall performance of the vehicle. Additionally, the results were evaluated based on several quality factors such as environmental impact, sustainability, safety, cost, and engineering standards.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

The key systems such as steering system, braking system and transmission system were modelled using CAD software and understood their arrangement. The analytical calculations were done to find out the performance parameters such as motor torque, tractive force, acceleration, and vehicle speed. This study gave a clear understanding of drivetrain and the influence of transmission on the performance of the go-kart.

6.2 Future Scope of Work

The present work can be further extended by performing detailed vehicle dynamics analysis to know the behaviour of the electric go-kart under different driving conditions. Future research can also focus on optimizing the transmission ratio and drivetrain configuration to improve acceleration and top speed for competition.

In addition, advanced simulations and structural analysis of drivetrain components can be carried out to improve reliability and performance. Further studies may also explore different motor specifications, battery capacities, and control systems to improve the overall efficiency of the electric go-kart. These improvements can help in developing more competitive and efficient electric go-karts for kart racing and student motorsport competitions.

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